

AWS Auto Scaling and Load Balancing

1. Protocols and Ports

Below are commonly used protocols with their standard ports:

- SSH – Secure Login Protocol – Port 22
- HTTP – Unsecure Web Protocol – Port 80
- HTTPS – Secure Web Protocol – Port 443

2. Auto Scaling Setup

Step-by-step instructions to configure Auto Scaling in a private network:

1. Create a launch template
2. Create an Auto Scaling group in the private subnet
3. Spin up two EC2 servers
4. Install Apache Web Server (httpd) using:
yum install httpd -y
5. Add a sample HTML page to the document root:

```
vim /var/www/html/index.html
```

```
<html>
<body>
<h1>This is the First Server</h1>
</body>
</html>
```

6. Restart the httpd service

3. Load Balancer Setup

Follow these steps to configure Application Load Balancer:

1. Create a target group
2. Create an Application Load Balancer (ALB) in the public subnet

3. Map the ALB with the Auto Scaling group
4. Verify that incoming traffic is routed evenly across the instances

4. Auto Scaling Concepts in AWS

Auto Scaling in AWS enables automatic adjustment of the number of EC2 instances based on demand. There are two types of scaling: Horizontal and Vertical.

4.1 Horizontal Auto Scaling

Horizontal scaling involves increasing or decreasing the number of EC2 instances based on load metrics such as CPU utilization or network traffic.

- Scaling Out – Launching additional instances when demand increases.
- Scaling In – Terminating instances when demand decreases.

4.2 Vertical Auto Scaling

Vertical scaling means resizing an instance (changing its type) based on performance requirements.

- Scaling Up – Upgrading instance size when performance demand increases.
- Scaling Down – Downgrading instance size to save cost during low demand.

5. Architecture: Request Flow & Access Control

End-user Request → Load Balancer → Target Groups → Auto Scaling EC2 Instances
(Private Web Servers)

Application servers (web or app servers) are hosted in private subnets for security.

To access private instances, a Bastion (Jump/Proxy) Host is deployed in the public subnet. SSH access is allowed only to the Bastion Host, which can then connect to private instances.